

WASP-43b

R-0612

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180312 · created 2026-07-25T08:23:33+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458189.743 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.018
Transit depth (Rp/Rs)^2	3.87 +/- 0.69 [%]
Orbital Inclination (inc)	86.96 +/- 1.57
Airmass coefficient 1 (a1)	1.116 +/- 0.026
Airmass coefficient 2 (a2)	-0.07 +/- 0.041
Scatter in the residuals of the lightcurve fit is	2.32 %
Best Comparison Star	#3 - [372, 45]
Optimal Method	PSF photometry
Transit Duration (day)	0.0622 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.197 ± 0.018 vs 0.1594 (deviation 1.88σ)
Duration fit vs published	0.0622 d vs 0.0512 d (deviation 5.18σ)
Mid-time O–C	0.9 min
Depth SNR	9.8
Residual scatter	2.32 %

Light curve

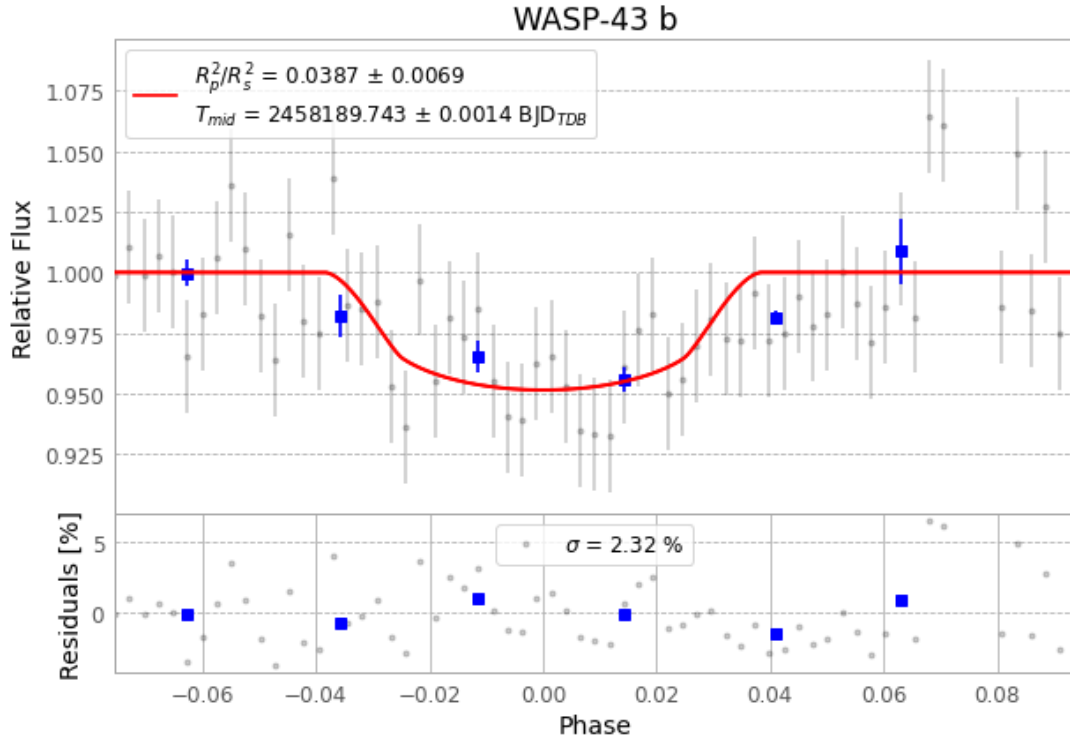


Figure 1 — FinalLightCurve_WASP-43 b_2018-03-11.png (SHA-256 82aaf823cd8d601b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [449, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f019b400ac1a8e70...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

67 FITS frames (44 MB), WASP-43180312041212.FITS → WASP-43180312073012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180312.log (b87e747f083f...), FinalLightCurve_WASP-43 b_2018-03-11.png (82aaf823cd8d...), FinalLightCurve_WASP-43 b_2018-03-11.csv (994a888ba8fe...)

Compute resources

Wall time 285.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 08:23Z.